Year 9 · Chapter 10 · Surface area and volume extension: answers (teacher copy)

10.6 Circumference and arc length (booklet p. 2)

1. **a** 56.5 cm ($2 \times \pi \times 9$)
   b 80° ($12.6 \div 56.5 \times 360° = 80.2°$)
   c 30.6 cm ($12.6 + 9 + 9$)

2. **Review 10.5**
   h = 6 cm ($54 = \frac{1}{2} \times (11 + 7) \times h$, so $h = 108 \div 18$)

3. **a** 63.7 m (the two ends make one circle: $\pi d = 400 - 200 = 200$)
   b 7.5 m ($\pi \times 2.4$: the straights do not change)

4. **a** 31.4 cm (20 minutes is $\frac{1}{3}$ of a turn: $\frac{1}{3} \times 2 \times \pi \times 15$)
   b 22.6 m (24 turns: $24 \times 94.2 = 2\,261.9$ cm)

5. **a** 50.3 cm ($\frac{1}{2} \times \pi \times 16 + 2 \times \frac{1}{2} \times \pi \times 8 = 8\pi + 8\pi$)
   b $\pi \times 16 = 50.3$ cm: the two small curves together are as long as the big one.

6. **a** 15.9 cm (the radius grows by $100 \div (2\pi)$)
   b The same: the gap is always $100 \div (2\pi) = 15.9$ cm, whatever the size of the circle.

10.7 Areas of circles and sectors (booklet p. 3)

1. **a** 44.8 m² ($\pi \times 5.5^2 - \pi \times 4^2$: the big circle has radius 4 + 1.5 = 5.5 m)
   b $\pi \times 1.5^2$ is a circle of radius 1.5 m (only 7.1 m²); the path is the big circle minus the pond.

2. **a** 21.5 cm² ($100 - 4 \times \frac{1}{4} \times \pi \times 5^2 = 100 - 25\pi$)
   b 21.5% (21.5 out of 100 cm²)

3. **a** large 706.9 cm², small 314.2 cm²
   b The large: it costs 2.5 cents per cm², the small 2.9 cents per cm² (or: it has 2.25 times the area for only twice the price).

4. **Review 10.6**
   a 7.0 cm ($\pi r + 2r = 36$, so $r = 36 \div (\pi + 2)$)
   b 77.0 cm² ($\frac{1}{2} \times \pi \times 7.0^2$)

5. **Review 10.4**
   a 46 m² (the outside is 14 m by 11 m: 154 − 108)
   b $2530 ($46 \times \$55$)

6. **a** 39.3 m² (a semicircle: $\frac{1}{2} \times \pi \times 5^2$)
   b 58.9 m² ($\frac{3}{4} \times \pi \times 5^2$)

10.8 Surface area of a prism (booklet p. 4)

1. **a** s = 2.5 m ($\sqrt{(1.5^2 + 2^2)}$)
   b 38 m² (two ends $2 \times \frac{1}{2} \times 3 \times 2 = 6$, then $4 \times (2.5 + 2.5 + 3) = 32$)

2. **a** 36 cm² ($8 \times 6 - 3 \times 4$)
   b 28 cm (the top is 3 + 5 = 8 cm and the right side 2 + 4 = 6 cm)
   c 184 cm² ($2 \times 36 + 28 \times 4$)

3. **Review 10.7**
   10.3 cm² (quarter circle $\frac{1}{4} \times \pi \times 6^2 = 28.3$, minus the triangle $\frac{1}{2} \times 6 \times 6 = 18$)

4. **a** both 24 cm³
   b A 52 cm², B 68 cm²
   c Box A: it holds the same amount but uses 16 cm² less card, so it costs less to make.

5. **a** 20 cm × 10 cm × 5 cm
   b 500 cm² ($200 + 2 \times 100 + 2 \times 50$)
   c It is the card less the four squares: $30 \times 20 - 4 \times 5^2 = 500$ cm².

6. **a** 65 m² (walls $2 \times (5 + 4) \times 2.7 = 48.6$, ceiling 20, less the door 1.8 and the window 1.8)
   b 3 tins (two coats: 130 m² ÷ 12 = 10.8 L)

10.9 Surface area of a cylinder (booklet p. 5)

1. **a** 12.0 cm ($301.6 \div (2 \times \pi \times 4)$)
   b 402.1 cm² ($301.6 + 2 \times \pi \times 4^2$)

2. **a** It wraps round the circles, so its length is their circumference.
   b 6.0 cm ($37.7 \div (2\pi)$)
   c 603.2 cm² ($37.7 \times 10 + 2 \times \pi \times 6.0^2$)

3. **a** both $2 \times \pi \times 5 \times 10 = 2 \times \pi \times 10 \times 5 = 314.2$ cm²
   b Q, by 471.2 cm² (P 471.2 cm², Q 942.5 cm²)
   c The rectangles are the same; Q's two circles are much bigger.

4. **a** 2.5 m² (40 cm = 0.4 m: $\frac{1}{2} \times 2 \times \pi \times 0.4 \times 2$)
   b 3.0 m² (the two half circles make one circle: $\pi \times 0.4^2 = 0.50$)

5. **Review 10.8**
   a 10 cm ($\sqrt{(8^2 + 6^2)}$)
   b 336 cm² ($2 \times \frac{1}{2} \times 8 \times 6 + 12 \times (8 + 6 + 10)$)

6. **a** 1 570.8 cm² ($\pi \times 10 \times 50$)
   b 1 256.6 cm² ($\pi \times 8 \times 50$)
   c 2 884.0 cm² (the ends: $2 \times (\pi \times 5^2 - \pi \times 4^2) = 56.5$)

10.10 Volumes of prisms and cylinders

(booklet p. 6)

1
- **a** 12.7 cm ($1000 \div (\pi \times 5^2)$)
- **b** 25.5 cm (twice as tall)

2
- **a** 502.7 cm^3
- **b** 1 005.3 cm^3 (r = 8, h = 5)
- **c** It doubles: doubling r makes r^2 four times as big, and halving h halves that.

3
- **a** 43.75 m^2 ($\frac{1}{2} \times (2.5 + 1) \times 25$)
- **b** 437 500 L (437.5 m^3)
- **c** 29 h 10 min (437 500 ÷ 250 = 1 750 minutes)

4 **Review 10.9**
- **a** 7.6 cm (the volume stays 216 cm^3: $h = 216 \div (\pi \times 3^2)$)
- **b** No: the cube's is 216 cm^2, the cylinder's 200.5 cm^2.

5 **Review 10.1**
- **a** 1 440 L (15 mm = 0.015 m: $12 \times 8 \times 0.015 = 1.44$ m^3)
- **b** 45.8 cm ($1.44 \div (\pi \times 1^2) = 0.458$ m)

6 **Review 10.7**
- **a** 74.1 cm^2 ($10 \times 6 + \frac{1}{2} \times \pi \times 3^2$)
- **b** 1 482.7 cm^3 (74.1×20)